



Verification Report:

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China Environmental United Certification Center Co., Ltd.	07 January 2010
Report Title:	Approved by:
Verification Report of Guangdong Lankou 26MW Hydro Power Project	Mr. TANG Dingding
Client:	Project Title:
KOE Environmental Consultancy Inc.	Guangdong Lankou 26MW Hydro Power Project
Summary:	
KOE Environmental Consultancy Inc. has commissioned China Environmental United Certification Center Co., Ltd. to carry out the verification to the project "Guangdong Lankou 26MW Hydro Power Project" with regard to the relevant requirements of VCS 2007.1 Standard. The verification scope includes three phases: 1) desk review of the project design and the baseline and monitoring plan; 2) on-site visit and follow-up interviews with project stakeholders; 3) resolution of outstanding issues and the issuance of the final verification report and opinion. The overall verification, from contract review to verification report and opinion, was conducted using CEC internal procedures. The first output of the verification process is a list of Corrective Action Requests and Clarification Requests. All CARs and CLs were successfully closed, the conclusions are presented in Section 3. CEC confirms that the project is implemented in accordance with the validated and registered CDM PDD. Installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately. The monitoring system is in place and the emission reductions are calculated without material misstatements. Our opinion relates to the project's GHG emissions and resulting GHG emissions reductions reported and related to the valid and registered project baseline and monitoring, and its associated documents. Based on the information we have seen and evaluated we confirm the following statement: Reporting period: 01/11/2007 to 06/07/2009 Baseline emissions: 82,009 tCO_{2eq} Project emissions: 0 tCO_{2eq} Emission reductions: 82,009 tCO_{2eq}	
Work carried out by:	Number of pages:
LIU Yaotian, YIN Yun	14

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1 Introduction

KOE Environmental Consultancy Inc. has commissioned China Environmental United Certification Center Co., Ltd. (CEC) to verify the emission reductions of its voluntary emission reduction (VER) project of Guangdong Lankou 26MW Hydro Power Project (hereafter referred to as “the project”), located in Bainitang village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, China for the period from 01/11/2007 to 06/07/2009 prior to its CDM first crediting period.

The report was carried out as per Voluntary Carbon Standard (VCS) version 2007. The report contains the findings from the verification and a statement for the verified emission reductions.

1.1 Objective

Verification is the periodic independent review and ex post determination by the verification body of the monitored reductions in GHG emissions during defined verification period.

The objective of verification is to verify the project is implemented as described in the project design document, and then to confirm the monitoring system is implemented and fully function to generate voluntary carbon units (VCUs) before its CDM first crediting period as from 07/07/2009 and after the time limit of project start day 01/11/2007 as prescribed in Clause 5.2.1 of VCS 2007.1 Program.

The verification also aims to confirm that the data reported are accurate, complete, consistent, transparent and free of material error or omission by checking the monitoring records and the emission reduction calculation.

1.2 Scope and Criteria

The verification scope is defined as an independent and objective review of the registered CDM PDD, VCS additional information, monitoring report, supporting documents made available to the auditor and information collected through performing interviews and during the on-site assessment. Furthermore publicly available information was considered as far as available and required.

CEC has employed a risk-based approach in the verification. The principles of accuracy and completeness, relevance, reliability and credibility were combined with a conservative approach to establish a traceable and transparent verification opinion.

The verification is not meant to provide any consultancy towards the client. However, stated requests for forward actions and/or corrective actions may provide input for improvement of the project monitoring towards reductions in the GHG emissions.

1.3 VCS project Description

Title of Project Activity:	Guangdong Lankou 26MW Hydro Power Project
Verification Period:	01/11/2007 to 06/07/2009
Project Starting Date:	01/11/2007

Location of the Project Activity: Bainitang Village, Lankou Town, Dongyuan County, Heyuan City, Guangdong Province, China

The project involves the construction of a hydro power station with a new reservoir. The project is located on the Dongjiang River which is one of the major four rivers of Zhu Jiang. The total installed capacity of the Project will be 26 MW (3*8.66 MW) with the power density of 183.82W/m². The designed annual operating hours are 3462h and annual net electricity generation is 80,070MWh which will displace part of the electricity generated by China Southern Power Grid (CSPG). CSPG is dominated by fossil fuel-fired power plants and thus greenhouse gas (GHG) emission reductions can be achieved. The implementation of the project is found to be in accordance with the registered CDM PDD.

During the verification period from 01/11/2007 to 06/07/2009, the total net electricity supplied by the project amounted to 94,133.8 MWh, and the GHG emission reduction is 82,009tCO₂e.

1.4 Level of assurance

The verification report is based on the registered CDM PDD, VCS additional information, monitoring report, supporting documents listed as references made available to the auditor and information collected through performing interviews and during the on-site assessment. The verification conclusion is assured the credibility of all above.

2 Methodology

Standard auditing techniques have been adopted. The verification team performs first a desk review, followed by an on-site visit and follow-up interviews with relevant project stakeholders. The next step is to close out the findings through direct communication with the PPs and finally prepare the verification report. This verification report and other supporting documents then undergo an internal quality control by the relevant technical review department of CEC before submission to issuance.

The criteria of this verification include the relevant rules and steps as set out in VCS 2007.1.

The overall verification process, from contract review to verification report and opinion, was conducted using CEC internal procedures.

Verification Team

Name	Role	Qualification
YIN Yun	Team Leader	Auditor of Specific Scope
LIU Yaotian	Team Member	Auditor Trainee

2.1 Review of Documents

The Monitoring Report (MR), Balance Advice Note (BAN) and Electricity-sale Invoice (ESI) were assessed as a part of the verification. In particular the baseline estimations and the monitoring plan contained in the registered CDM PDD were also assessed.

2.2 On-site visit

In the period of 23-25 November 2009, CEC performed a site-visit at Guangdong Lankou 26MW Hydro Power Project in Guangdong Province. During this visit CEC:

- Confirm the implementation and operation of the project;
- Resolve issues identified in the document review;
- Confirm the correct implementation of procedures for operations and data collection;
- Cross-check the information provided in the MR documentation with other sources (raw data);
- Check the monitoring equipments against the requirements of the CDM PDD and the approved methodology, including calibrations, maintenance, etc.;
- Identify if the quality control and quality assurance procedures are in place to prevent or correct errors or omissions in the reported parameters.

2.3 Reporting of Findings

As an outcome of the verification process, the verification team can raise different types of findings. Findings established during the initial verification could either be seen as a non-fulfillment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

Corrective Action Request (CAR) is issued, where:

- Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- Issues identified in a FAR during validation to be verified during verification have not been resolved by the project participants.

Clarification Request (CL) is issued, where:

- Information is insufficient or not clear enough to determine whether the applicable CDM requirements have been met.

Forward Action Request (FAR) is issued, where:

- The monitoring and reporting require attention and/or adjustment for the next verification period.

2.4 Internal Quality Control

As an ultimate step of verification, the final documentation including the verification report is reviewed by technical reviewers and approved by the CDM Supervisor. The technical reviewers are not part of the verification team, the review is conducted independently of the verification team.

3 Verification Findings

In the following sections, the findings of the verification are stated. The verification findings for each verification subject are presented as follows:

The findings from the desk review of the original project activity documents and the findings from interviews during the follow up visit are summarized. The verification of the project resulted in **1 Corrective Action Request and 2 Clarification Requests**.

The conclusions for verification subject are presented.

In the final verification report, the discussions and the conclusions that followed the preliminary verification report and possible corrective action requests should also be summarized in this section.

3.1 Remaining issues, including any material discrepancy, from previous validation

There is no remaining issue from previous validation. The verification has been carried out based on the monitoring report, CDM PDD, initial verification and supporting documents provided.

3.2 Project Implementation

The project was implemented and the equipment was installed as described in the registered CDM PDD. There are no major changes in the key equipment since the validation of the project. All 3 sets of hydro-turbine generators in the power plant with the total installed capacity of 26MW have been put into operation since from 01/11/2007. For this verification period, the electricity generated from 01/11/2007 to 06/07/2009 was taken into consideration.

The project exported electricity to the China Southern Power Grid (CSPG) via the Lankou 110kV Transformer Station. The recorded generation data, meter readings, meter calibration certificates, electricity-sale invoices, balance advice note and plant operation records were verified by the verification team during the on site visit.

Verification team has verified the nameplates of the turbines as GZ1250a-WP-550 and the generators as SFWG7350-80/6220.

For the metering purpose, a main meter is installed for measuring the net electricity delivered to the grid at the substation as the gateway meter, furthermore, a backup meter is installed at the exit power plant. Both of these meters are installed and calibrated in accordance with the registered CDM PDD. The calibration of meters as follows:

	Main meter	Back-up meter
Type/Model	DTSD341	YD2103
SN	01010400497	070817234
Precision	0.5s	0.5s
Calibration Periods	12/04/2007	12/04/2007
	12/04/2008	12/04/2008

	12/04/2009	12/04/2009
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3.3 Completeness of Monitoring

CEC has verified the reported data and procedures actually implemented by the PP during on site verification and confirmed that:

- The monitoring plan is in accordance with the approved methodology ACM0002 version 07, applied by the proposed CDM project activity.
- The monitoring has been carried out in accordance with the monitoring plan contained in the registered CDM PDD. All parameters were monitored and determined as per the monitoring plan.
- The documented evidences and records submitted by PP to determine emission reductions have been verified as of sufficient quantity and appropriate quality.

3.4 Accuracy of Emission Reduction Calculations

The calculation of emission reductions has been verified against the documented evidences on-site and presented by the PP and found accurate.

According to the registered monitoring plan and the monitoring report, the following internal parameters need to be monitored for determining the Net electricity exported to the grid:

$EG_{output, y}$ Electricity delivered to the grid

$EG_{input, y}$ Electricity used by the proposed project bought from the power grid

The electricity exported to and imported from the grid is monitored by the gateway metres installed at the substation which is recorded by the Grid Company.

The net electricity (EG_y) is calculated as the following equation:

$$EG_y = EG_{output, y} - EG_{input, y}$$

The reported data of the monitoring report is generated according to the reading records of the main meter. According to the monitoring plan, the emission reductions are to be calculated using the net electricity delivered to the grid measured by the main meter and the data of the meter shall be cross checked against the BAN and the ESI. CEC has checked the records of main meter, the BAN and the ESI on site and found all of them are consistent.

According to the methodology ACM0002 version 07, there are no other external data are required for determining the emission reductions of the project as the emission factor of the project has been determined ex-ante in the registered CDM PDD. The emission factor used in the monitoring report has been verified against the CDM PDD and found the same.

3.5 Quality of Evidence to Determine Emission Reductions

The data and parameters need to be monitored during the reporting period are the electricity exported to and imported from the grid, the total electricity produced by the project activity, the installed capacity of the hydro power plant after the implementation of the project activity and the surface area at full reservoir level.

The key parameters used for the determination of the emission reductions are the reading records of the main meter for electricity exported to and imported from the grid. CEC has taken a specific cross check for all the data via the further sources available including the raw data. These data are maintained in the identified records and in compliance with the Summary of Emission Reduction Calculation Form.

As per the documents review and on-site visit, CEC confirmed that the total electricity produced by the project activity is higher than the electricity exported to the grid because of the line loss, the installed capacity of the hydro power plant after the implementation of the project activity is 26MW, which was verified by the nameplates of the turbines and generators, it is consistent with the registered CDM PDD. CEC also verified that the surface area of the reservoir when it is full per year during the reporting period does not exceed the measured value in the registered CDM PDD (141.441m²).

According to the registered CDM PDD and the monitoring report, the emission reductions for the project are calculated as the baseline emissions minus the project emissions and leakage.

According to the measuring result per year during the reporting period, the power density of the Project is 183.82W/m², greater than 10W/m², therefore, the project emissions is zero.

The leakage need not be considered as per the registered CDM PDD and the methodology ACM0002 version 07.

The final GHG emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - L_y = BE_y - 0 - 0 = EG_y \times EF_y$$

Where:

ER_y Emission reductions

BE_y Baseline emissions

PE_y Project emissions

L_y Leakage

EG_y Net electricity supplied to the grid

EF_y GHG emission factor of the China Southern Power Grid (CSPG), calculated ex-ante and will not be updated during this period, the value is 0.8712tCO₂e/MWh.

According to the registered monitoring plan, data measured by the meter will be cross checked against the BAN and the ESI. CEC has checked the Monthly Reading Records (MRR) of main meter, the BAN and the ESI, and found they are consistent.

The verified electricity indicated in the MR and Emission Reduction Calculation Form has been cross checked with the MRR, BAN and ESI as follows:

Period	EG _{output, y} (MWh)	EG _{input, y} (MWh)	EG _y (MWh)
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	A	B	C=A-B
01/11/2007-30/11/2007	1127.909	0.000	1127.909
01/12/2007-31/12/2007	5580.000	0.000	5580.000
01/01/2008-31/01/2008	6690.000	0.000	6690.000
01/02/2008-29/02/2008	4460.000	0.000	4460.000
01/03/2008-31/03/2008	4460.000	0.000	4460.000
01/04/2008-30/04/2008	4460.000	0.000	4460.000
01/05/2008-31/05/2008	1140.000	0.000	1140.000
01/06/2008-30/06/2008	3350.000	0.000	3350.000
01/07/2008-31/07/2008	2196.700	0.000	2196.700
01/08/2008-31/08/2008	9171.360	0.000	9171.360
01/09/2008-30/09/2008	6779.520	0.000	6779.520
01/10/2008-31/10/2008	5466.780	0.000	5466.780
01/11/2008-30/11/2008	4051.740	0.000	4051.740
01/12/2008-31/12/2008	3548.160	0.000	3548.160
01/01/2009-31/01/2009	3742.860	0.000	3742.860
01/02/2009-28/02/2009	3519.120	0.000	3519.120
01/03/2009-31/03/2009	4951.980	0.000	4951.980
01/04/2009-30/04/2009	5325.540	0.000	5325.540
01/05/2009-31/05/2009	4877.400	0.000	4877.400
01/06/2009-30/06/2009	7795.920	0.000	7795.920
01/07/2009-06/07/2009	1633.500	194.700*	1438.800
Total	94,328	194.700	94,133.8

*: As for the electricity imported from the Grid is a tiny number, the Grid company would not issue the BAN to the PP every month. This data is the total value of the Power purchased from the Grid from the period of 01/11/2007 to 06/07/2009, which could be cross-checked by the specific BAN issued by the Heyuan Grid Company for the electricity imported from the Grid.

The emission reductions in this verified period:

$$ER_y = EG_y \times EF_y = 94,133.8 \text{ MWh} \times 0.8712 \text{ tCO}_2\text{e/MWh} = 82,009 \text{ tCO}_2\text{e}$$

For 01/11/2007-31/12/2007, the emission reductions are:

$$ER_y = EG_y \times EF_y = 6,707.909 \text{ MWh} \times 0.8712 \text{ tCO}_2\text{e/MWh} = 5,844 \text{ tCO}_2\text{e}$$

For 01/01/2008-31/12/2008, the emission reductions are:

$$ER_y = EG_y \times EF_y = 55,774.260 \text{ MWh} \times 0.8712 \text{ tCO}_2\text{e/MWh} = 48,590 \text{ tCO}_2\text{e}$$

For 01/01/2009-06/07/2009, the emission reductions are:

$$ER_y = EG_y \times EF_y = 31,651.620 \text{ MWh} \times 0.8712 \text{ tCO}_2\text{e/MWh} = 27,575 \text{ tCO}_2\text{e}$$

CEC has confirmed that the calculation of the emission reductions is correct and conservative.

CEC has ensured that:

- The data used for the determination of the emission reductions are available and monitored in accordance with the registered monitoring plan without conservative assumption;
- The relevant commercial invoices (BAN, ESI) have been cross checked and found consistent;
- The appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been properly followed the methodology and registered CDM PDD;
- The emission factor of CSPG determined ex-ante in the registered CDM PDD is applied in the monitoring report and the calculations have been justified.

One CAR and two CLs were raised and successfully closed out.

CAR 01	
Description of finding	The emission reduction calculation process does not consider the electricity import from the Grid, which may lead the result to be not conservative. Moreover, during the on site visit, there is not any proof for the imported electricity that could be provided to the verification team. Please supply the relevant evidence and re-calculate the emission reduction.
Summary of project owner response	The PP has supplied the evidence of the data of the Power purchased from Grid company to the verification team and revised the emission reduction both in the VER sheet and the MR.
Conclusion	The specific BAN issued for the electricity imported from the Grid and the revised MR has been provided by the PP. CEC verified the relevant documents and confirms that the amount of the electricity import from the Grid is creditable and the re-calculated emission reduction is conservative. Hence CAR01 is closed.

CL 01	
Description of finding	Please supply the proof document of the surface area of the reservoir when it is full per year during the verification period.
Summary of project owner response	The PP has supplied the documents of the water level curve each year, and also the photographs of the reservoir at several key locations.

Conclusion	The documents of the water level curve of the reservoir and the photographs of the reservoir at several key locations in year 2007, 2008 and 2009 has been provided by the PP. CEC verified these proof documents and confirms that the surface area of the reservoir when it is full per year during the reporting period does not exceed the measured value in the registered CDM PDD (141.441m ²). Hence CL01 is closed.
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CL 02	
Description of finding	Some of the units of the parameters monitored in MR are inconsistent with the registered CDM PDD, further clarification is required.
Summary of project owner response	The PP has supplied the revised MR with the description of the units of the parameters monitored consistent with the registered CDM PDD.
Conclusion	The revised MR has been provided by the PP. CEC verified the updated MR and confirms that the revised description of the units of the parameters is consistent with the registered CDM PDD now. Hence CL02 is closed.

3.6 Management and Operational System

In order to verify data quality, mainly for the net electricity supplied to grid, operational and management structure was checked via interviewing relevant staffs and reviewing documents. CEC can confirm that the management system of the project is in place, with the responsibilities properly identified based on monitoring plan in registered CDM PDD. And the key parameters are measured and reviewed periodically as per the procedures.

4 Verification Conclusion

KOE Environmental Consultancy Inc. has commissioned China Environmental United Certification Center Co., Ltd. (CEC) to carry out the verification of the Guangdong Lankou 26MW Hydro Power Project, with regard to the relevant requirements of VCS 2007.1. The project reduces GHG emissions due to the generation of electricity by using of available hydro potential.

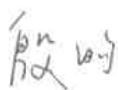
CEC verified the Monitoring Report and confirms that the project is implemented as planned and described in validated and registered CDM PDD. Installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately. The monitoring system is in place and is as per the approved methodology ACM0002 version 07.

In the course of the verification 1 CAR and 2 CLs were raised and successfully closed.

As the result of the verification, CEC confirms that the GHG emission reductions are calculated without material misstatements. Based on the information we have seen and evaluated, we confirm that the project has achieved emission reductions in the reporting period mentioned above as follows:

Period	Emission Reductions
01/11/2007-31/12/2007	5,844 tCO ₂ e (VCUs)
01/01/2008-31/12/2008	48,590 tCO ₂ e (VCUs)
01/01/2009-06/07/2009	27,575 tCO ₂ e (VCUs)
Total	82,009 tCO ₂ e (VCUs)

Beijing, 07/01/2010



YIN Yun
Verification Team Leader

Beijing, 07/01/2010



ZHANG Xiaodan
CDM Supervisor, Technical Director

5 References

Project related documents

- /1/ BAN of November 2007 to July 2009 issued by the Grid Company
- /2/ Business License of the PP
- /3/ CDM Monitoring and Management Manual
- /4/ Certificate of authorization for Heyuan Grid Company
- /5/ Daily/Monthly Reading Records of the meters
- /6/ Diagram of Power System Connection of the project
- /7/ Electricity-sale Invoice of November 2007 to July 2009 issued by the project participant
- /8/ Full Calibration Certificates of gateway meters and auxiliary meters
- /9/ Monitoring Report
- /10/ On-site Calibration Certificates of gateway meters and auxiliary meters
- /11/ Photographs of project site
- /12/ Staff operation certificates
- /13/ Signed Power Purchase Agreement (PPA) dated 20/08/2007
- /14/ Surface area at full reservoir level measurement result
- /14/ Technical particulars of turbines and generators
- /15/ Training records

Background investigation and assessment documents

- /1/ Approved CDM Methodology ACM0002 version 07
- /2/ Emission calculation sheets provided by the project participant
- /3/ National standard-“Technical Administrative Code of Electric Energy Metering DL/T 448-2000”
- /4/ Project Design Document version 5.0 dated 01/03/2009
- /5/ Validation and Verification Manual version 01.1 dated 04/12/2009
- /6/ Validation report for the project Guangdong Lankou 26MW Hydro Power Project

Websites used

- /1/ UNFCCC website <http://cdm.unfccc.int>
- /2/ VCS website <http://www.v-c-s.org>

Persons interviewed

Ms. ZHAO Xuejiao, Project Manager, KOE Environmental Consultancy Inc.



Ms. ZHANG Chang, Project Manager, KOE Environmental Consultancy Inc.

Mr. CHEN Dehua, Power Plant Master, Shenzhen City Fuyuan Industry Group Co., Ltd.

Mr. HE Jiandong, Accountant, Shenzhen City Fuyuan Industry Group Co., Ltd.

Mr. CHEN Bin, Engineer, Shenzhen City Fuyuan Industry Group Co., Ltd.